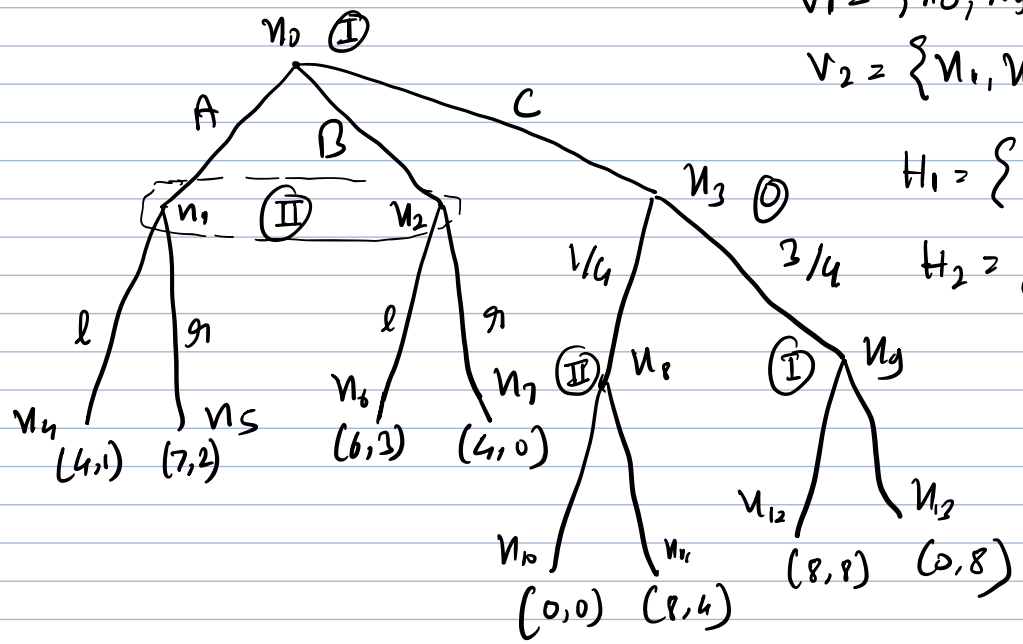
Class 23



$V_1 = \{n_0, n_9\}$

$V_2 = \{n_1, n_3, n_8\}$

$H_1 = \{\{n_0\}, \{n_3\}\}$

$H_2 = \{\{n_1, n_2\}, n_8\}$

$A(h)$ $\quad h = \{n_1, n_2\}$

$$A(h) = \{\{\{n_1, n_4\}, \{n_2, n_6\}\}, \{\{n_1, n_5\}, \{n_2, n_7\}\}\}$$

A pure strategy for player i, s_i, is a mapping

$$s_i : H_i \to \bigcup_{h \in H_i} A(h) \quad \text{s.t.}$$

$$s_i(h) \in A(h) \quad \forall h \in H_i$$

S_i is the pure strategy set of player i

$S = S_1 \times S_2 \times \cdots \times S_n$

The utilities at strategy profiles are defined as the expected utilities.

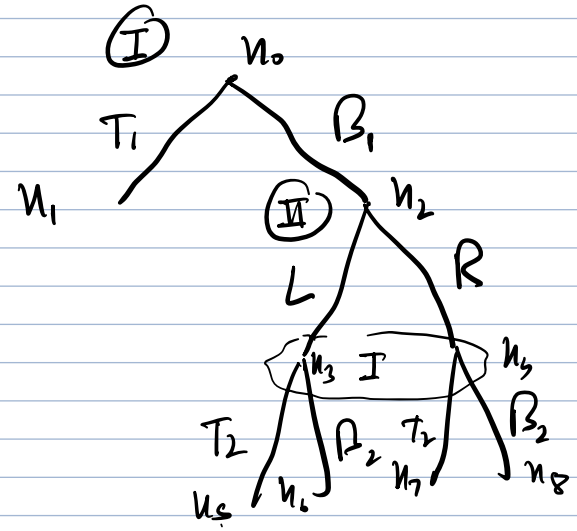
- Mixed strategies are prob distributions over pure

strategies.

For player i, the set of all mixed strategies is Σ_i

strategy-profile $\sigma = (\sigma_1, \ldots, \sigma_n)$

Mixed strategies are defined as convex combinations of mixtures of pure strategy profiles.



$S_1 = \{T_1 T_2, T_1 B_2, B_1 T_2, B_1 B_2\}$

$\Sigma_1 = \Delta_4$

$= \{(n_1, \ldots, n_4)$
$n_i \geq 0$,
and $\sum_{i=1} n_i = 1\}$

$$B_1 = \left\{ (x, 1-x), (y, 1-y) \;\middle|\; \begin{array}{l} 0 \leq x \leq 1 \\ 0 \leq y \leq 1 \end{array} \right\}$$

set of behavior strategies

- A behavior strategy $b_i : H_i \to \bigcup_{h \in H_i} \Delta A(h)$

s.t. $b_i(h) \in \Delta A(h)$
$\forall h \in H_i$

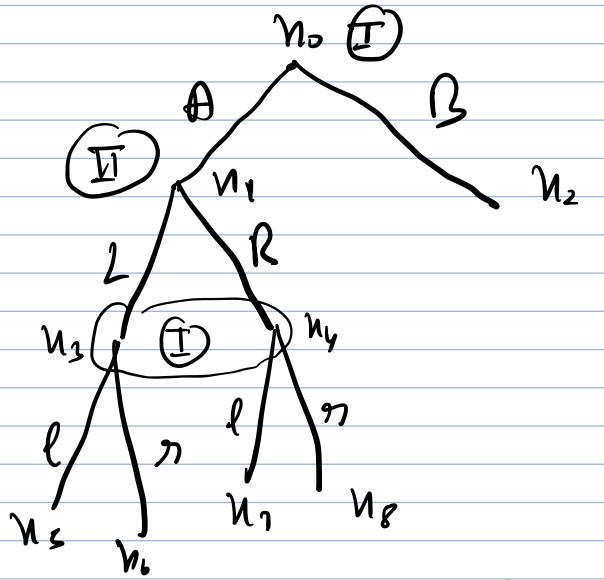
ΔS ! denotes a prob distribution over S.

- For a behavior strategy b_i, the prob that b_i will select action a at the information set h_i, $b_i(a, h_i)$

- $b = (b_1, \ldots, b_n)$ is a behavior strategy profile

- With a slight abuse of notations we denote $\sigma = (\sigma_i)_{i \in N}$ a strategy profile where σ_i can be either mixed or behavior strategy for player i.

- For a mixed / behavior strategy profile $\sigma = (\sigma_i)_{i \in N}$, $P(v; \sigma)$, where $v \in V$, denotes the prob that the node v will be reached when the underlying strategy profile is $\sigma = (\sigma_i)_{i \in N}$



$\sigma_1 = \left[\frac{2}{10}(Al), \frac{1}{10}(Ar), 4/10(Bl), 2/10(Br)\right]$

$\sigma_2 = \left[\frac{1}{2}(L), \frac{1}{2}(R)\right]$

v_5 will be reached with $3/10 \times \frac{1}{2} = \frac{3}{20}$

n_2 will be reach if $P-1$ plays either A_1 or A_3 and $P-2$ plays L_1
Ale prob is $[3/10 + 1/10] \times 1/2 = 4/20$

Equivalence Defn

A mixed strategy σ_i and a behaviour strategy b_i of Player i are equivalent if for every mixed/ behavior strategy vector σ_{-i} of the other players and every vertex $v \in V$.

$$P(v_j(\sigma_i, \sigma_{-i})) = P(v_j(b_i, \sigma_{-i}))$$